

CFD_HW_3

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课程：计算流体力学

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Question 11: Modified Equation & Stability Analysis

(a)

The code is :

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Program main
implicit none
! on wall implementation, center difference
! Euler time integration
!integer,parameter      :: N=32, Ntime=32
!real*8,parameter      :: dt = 0.0125d0
!integer,parameter      :: N=16, Ntime=8
!real*8,parameter      :: dt = 0.05d0
integer,parameter      :: N=16, Ntime=8
real*8,parameter      :: dt = 0.05d0

real*8,parameter      :: u0=1.0d0, aL=1.0d0, anu=0.1
! u0 maximum velocity,aL channel half width,anu kinematic viscosity
! body force is then g = 2*nu*u0/aL^2
!real*8, dimension(1:N+1) :: uold, u, uana [uana is not used, can be removed.]
real*8, dimension(1:N+1) :: uold, u
real*8                  :: theta0, theta1,
theta2,theory1,theory2,ss,sss,ycc,CFL,anu_m,tt
integer                 :: j,it,nsteps,k,output_count
real*8                  :: dy,pi,t,Tend

    pi = 4.0d0*atan(1.0d0)
    dy = 2.d0*aL/real(N)
! initial condition
    u = 0.0d0
    uold = u
    CFL = dt*anu/dy/dy

    write(*,*) 'anu, dt, aL, dy, CFL=', anu, dt, aL, dy, CFL
    t = 0.d0
    Tend = 5.1d0*aL*aL/anu
    nsteps = Tend / dt
    write(*,*) 'Tend, dt, Ntime, nsteps=', Tend, dt, Ntime, nsteps

    do it=1, nsteps

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do j=2,N
    u(j) = uold(j) + CFL*(uold(j-1)-2.d0*uold(j)+uold(j+1)) &
        + dt*2.d0*u0*anu/aL**2.d0
end do

do j=2,N
    uold(j) = u(j)
enddo

t = t + dt
! print out solutions
if(mod(it,Ntime).eq.0) then
! analytical solution with the same N
    output_count = output_count + 1
! assume N is even
    do j = 1,N+1
        ycc = real(j-1)*dy - 1.0d0
        theory1 = (1.d0- ycc**2.d0)
        theory2 = (1.d0- ycc**2.d0)
! I used 100 terms
        do k =0,100
            ss = (0.5d0+real(k))*pi

            sss = 2.d0 * ss / real(N)

            anu_m = anu * (1.0d0 + (CFL/2.d0 - 1.d0/12.d0) * sss**2.d0 &
                + (CFL**2.d0/3.0d0 - CFL/12.d0 + 1.0d0/360.0d0) * sss**4.d0)

            theta0 = 4.0d0*(-1.0)**k/(pi*(real(k)+0.5d0))**3.d0
            !
            theta1 =theta0 *exp(-ss*ss*anu*t)*cos(ss*ycc)
            theta2 =theta0 *exp(-ss*ss*anu_m*t)*cos(ss*ycc)

            theory1 = theory1 - theta1
            theory2 = theory2 - theta2
        end do

! Write down the solution at y =0
        if(abs(ycc) < 1.d-8) then
            tt = anu * (it * dt) / (aL**2)
            write(2,100) REAL(output_count),ycc,tt,abs(theory2-
theory1)/u0,abs(u(j)-theory1)/u0
        end if

        write(8,100) REAL(output_count),ycc,REAL(it),abs(theory1-
u(j))/u0,abs(theory2-u(j))/u0
    end do
    !!!!!!!!!!!

end if

100      format(2x, 5f16.12)

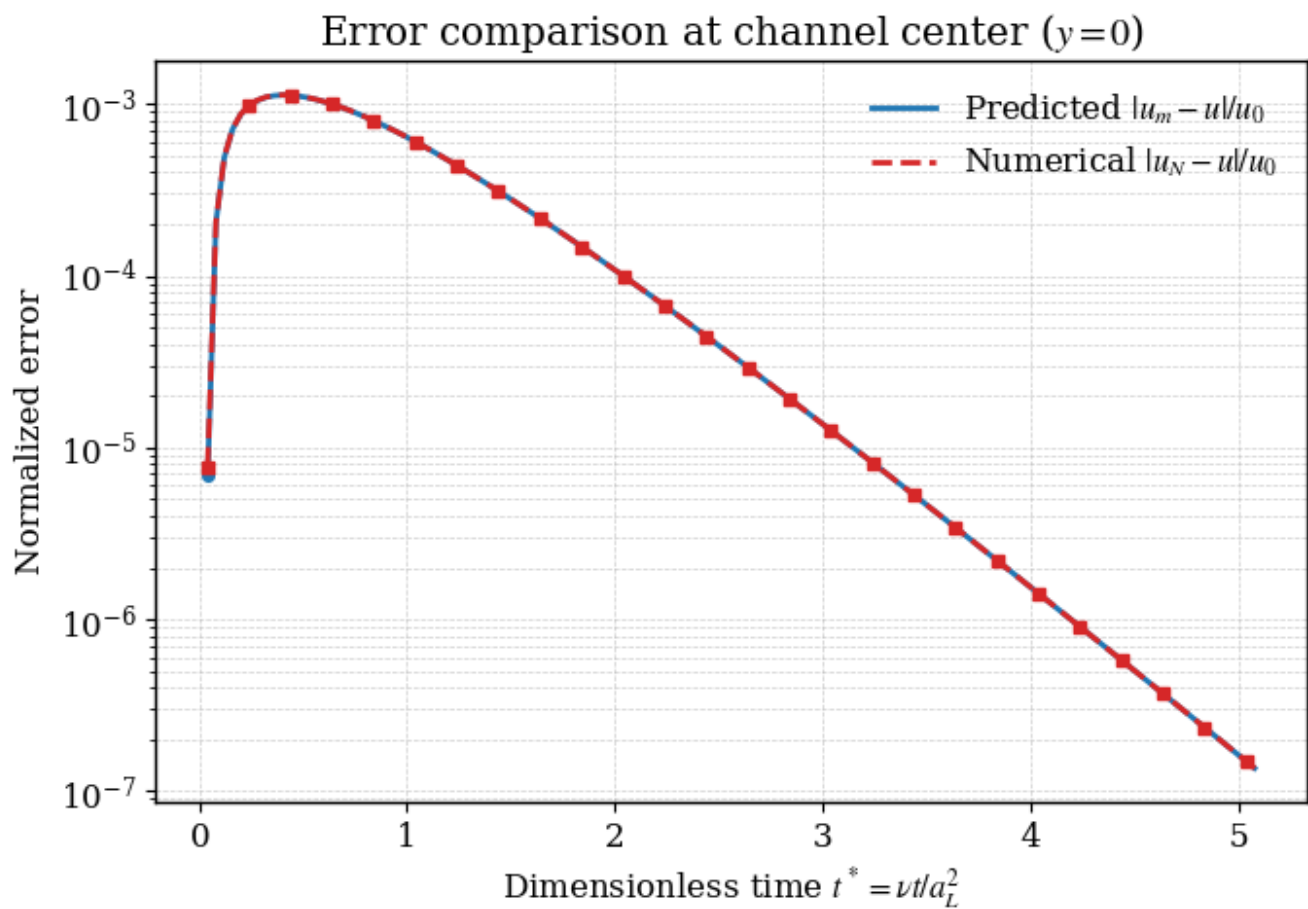
```

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enddo
end program

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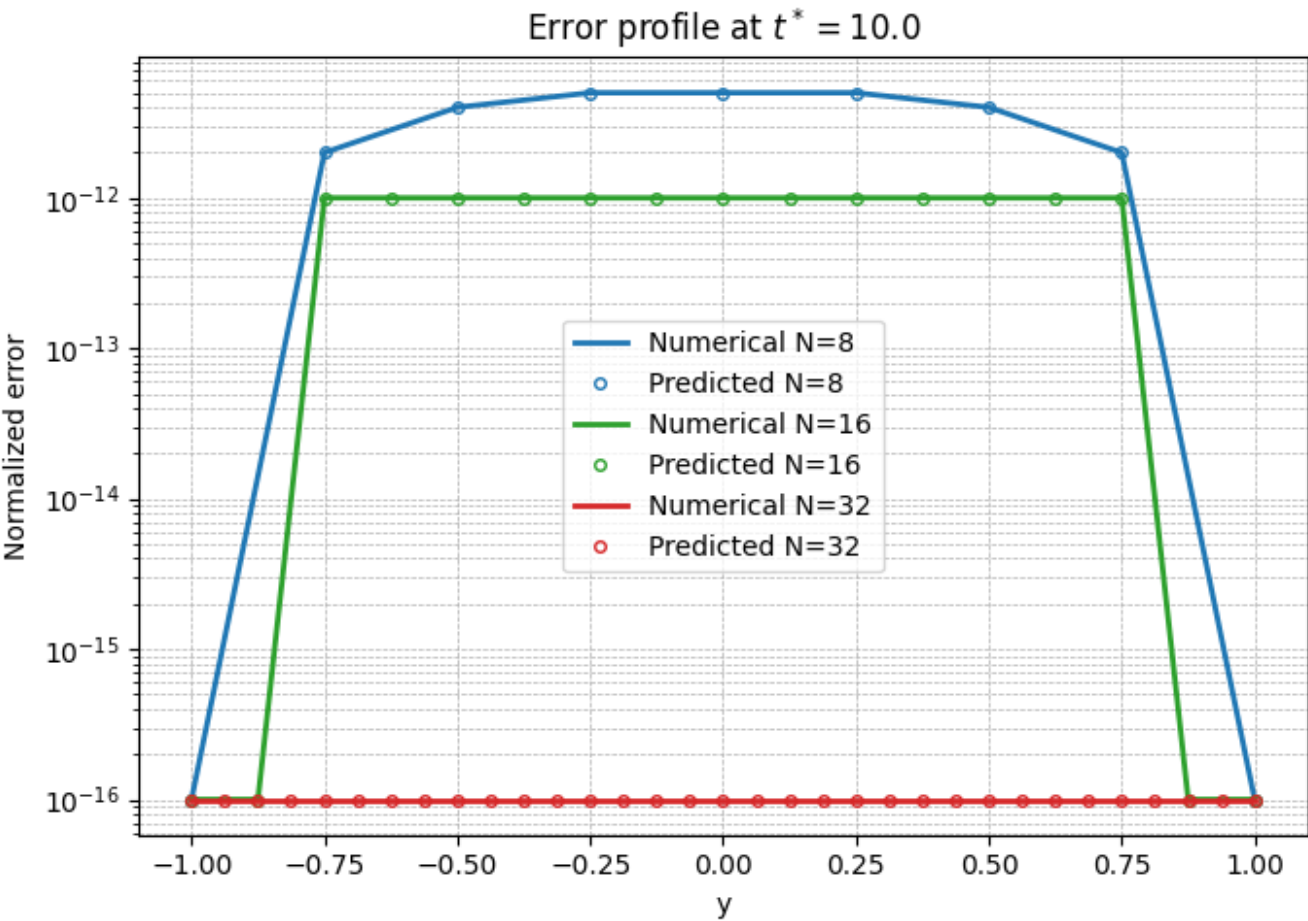
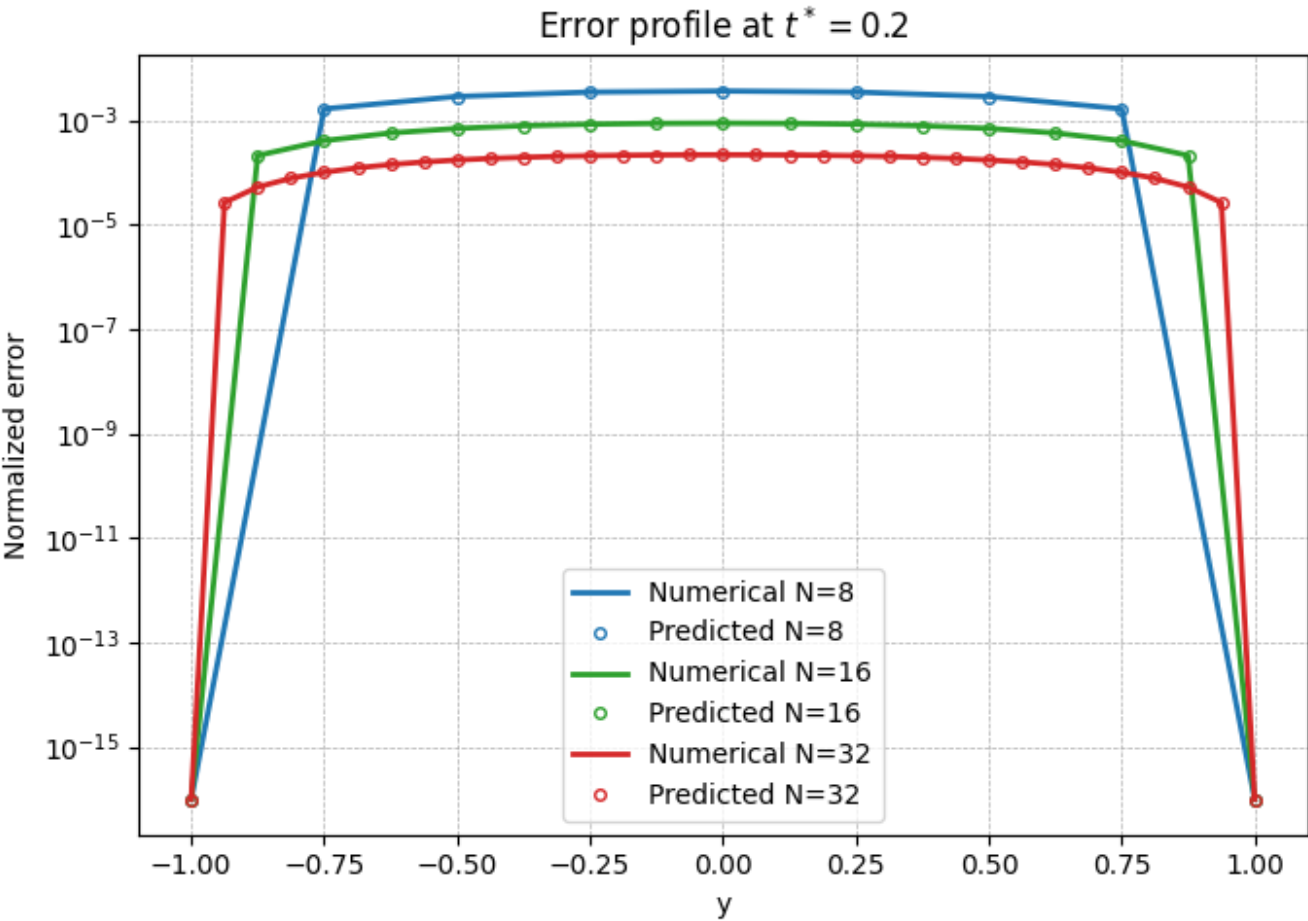
And the result is :



It shows that for 100 **ks**, the numerical solution and the analytical solution of the modified equation are almost the same.

(b)

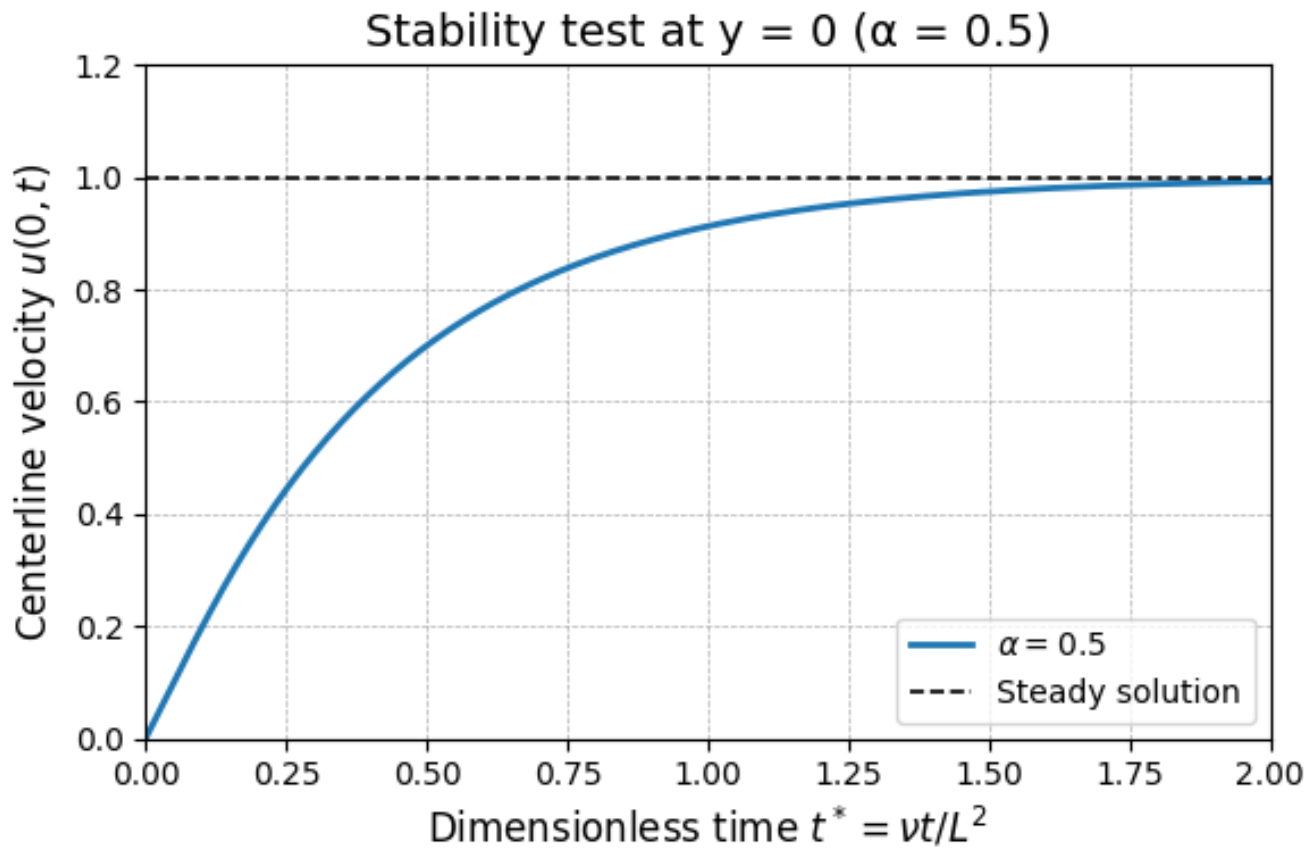
The results are :

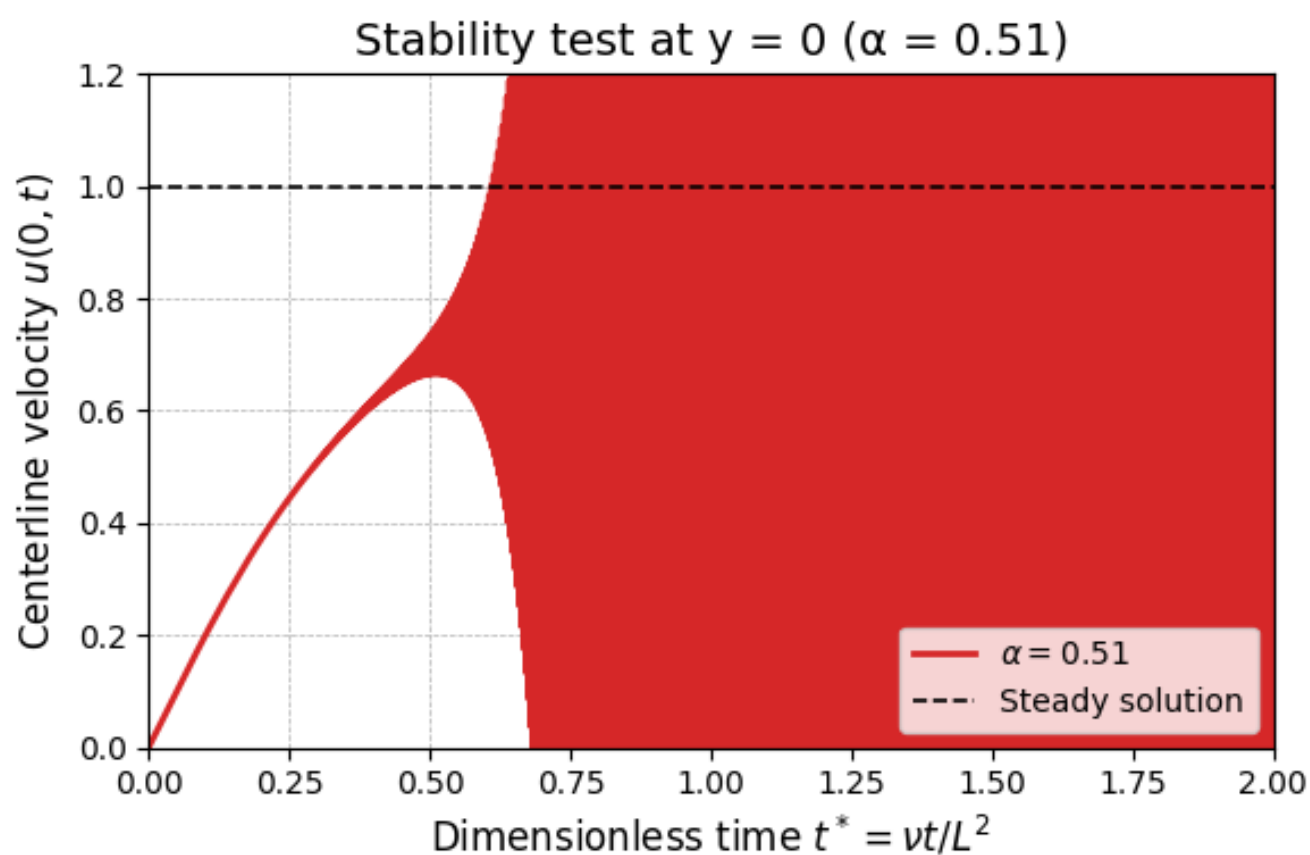
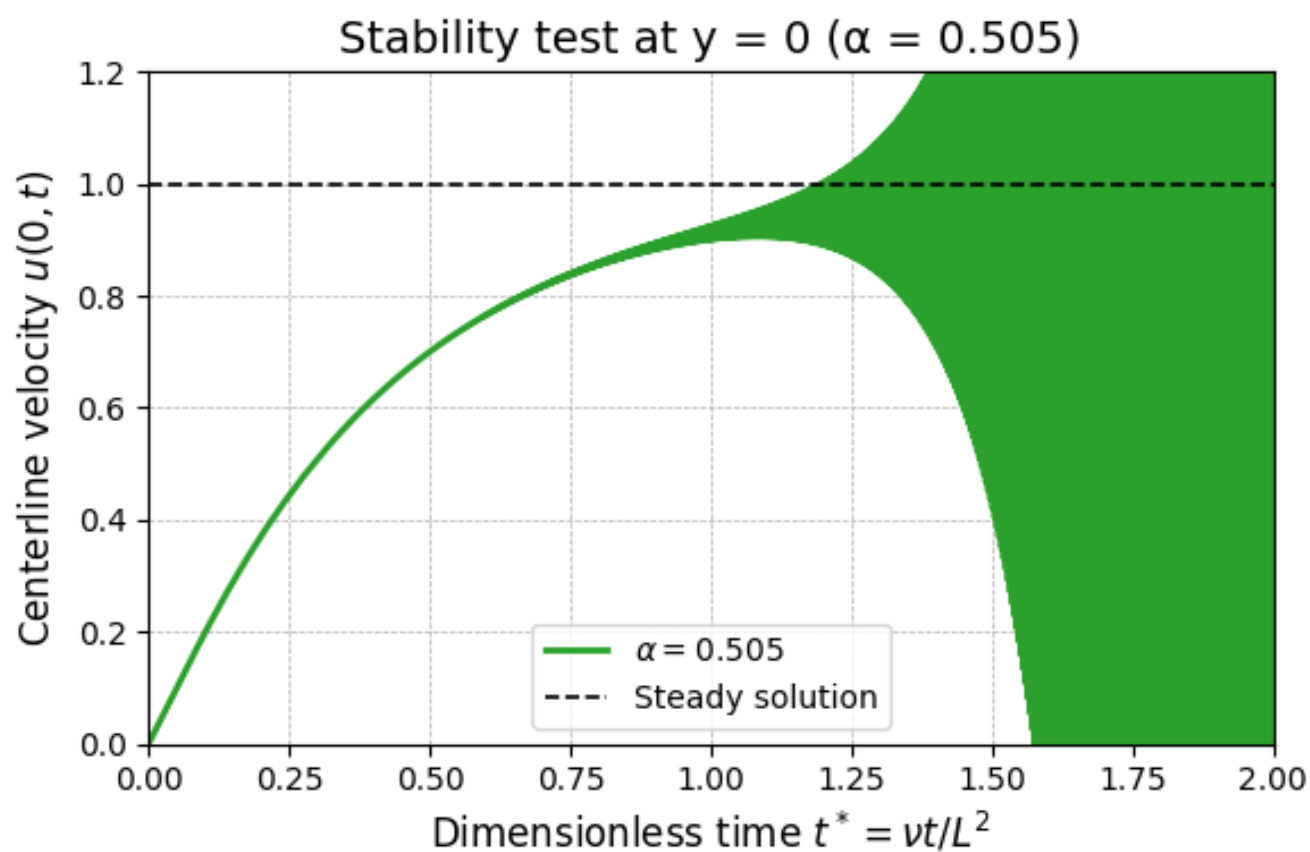


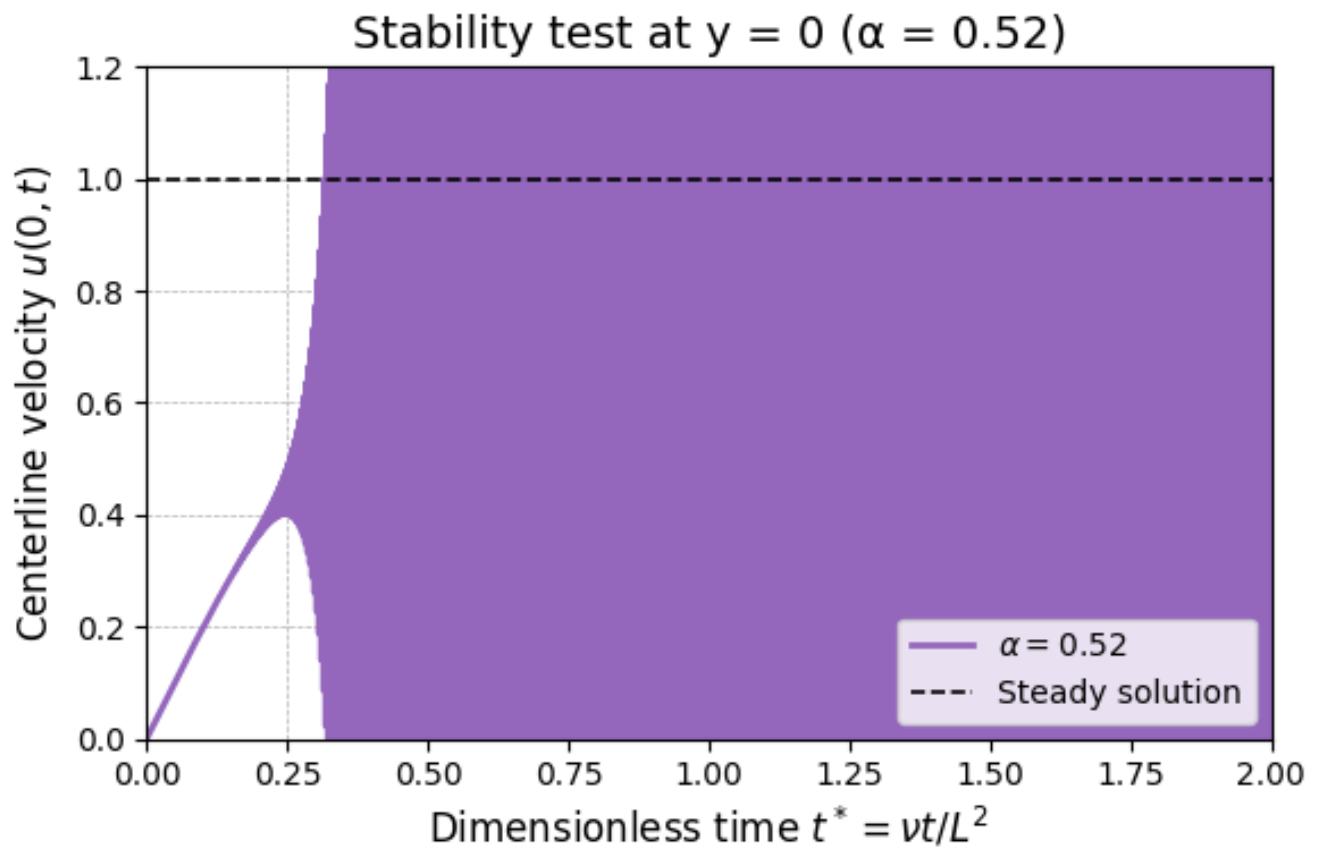
The predicted errors (markers) align closely with the numerical errors (solid lines), demonstrating that the modified equation accurately captures the numerical error distribution. And the smaller the grids are, the smaller the errors will be.

(c)

The results are :







If CFL=0.5 it will not diverge (in 2 dimensionless times), but as the CFL becomes larger, the divergence will happen earlier.

EOF

Q.9

$$\begin{aligned}
 a) \quad u_{i+1, j+1} &= u_{i,j} + (\Delta x) u_x + (\Delta y) u_y + \frac{1}{2!} \left[(\Delta x)^2 u_{xx} + (\Delta y)^2 u_{yy} + 2(\Delta x)(\Delta y) u_{xy} \right] \\
 &+ \frac{1}{3!} \left[(\Delta x)^3 u_{xxx} + (\Delta y)^3 u_{yyy} + 3(\Delta x)^2(\Delta y) u_{xxy} + 3(\Delta x)(\Delta y)^2 u_{xyy} \right] \\
 &+ \frac{1}{4!} \left[(\Delta x)^4 u_{xxxx} + (\Delta y)^4 u_{yyyy} + 4(\Delta x)^3(\Delta y) u_{xxx} + 6(\Delta x)^2(\Delta y)^2 u_{xxyy} + 4(\Delta x)(\Delta y)^3 u_{xyyy} \right] \\
 &+ O[(\Delta x)^5, (\Delta y)^5]
 \end{aligned}$$

$$\begin{aligned}
 u_{i-1, j-1} &= u_{i,j} - (\Delta x) u_x - (\Delta y) u_y + \frac{1}{2!} \left[(\Delta x)^2 u_{xx} + (\Delta y)^2 u_{yy} + 2(\Delta x)(\Delta y) u_{xy} \right] \\
 &+ \frac{1}{3!} \left[-(\Delta x)^3 u_{xxx} - (\Delta y)^3 u_{yyy} - 3(\Delta x)^2(\Delta y) u_{xxy} - 3(\Delta x)(\Delta y)^2 u_{xyy} \right] \\
 &+ \frac{1}{4!} \left[(\Delta x)^4 u_{xxxx} + (\Delta y)^4 u_{yyyy} + 4(\Delta x)^3(\Delta y) u_{xxx} + 6(\Delta x)^2(\Delta y)^2 u_{xxyy} + 4(\Delta x)(\Delta y)^3 u_{xyyy} \right] \\
 &+ O[(\Delta x)^5, (\Delta y)^5]
 \end{aligned}$$

$$\begin{aligned}
 u_{i-1, j+1} &= u_{i,j} - (\Delta x) u_x + (\Delta y) u_y + \frac{1}{2!} \left[(\Delta x)^2 u_{xx} + (\Delta y)^2 u_{yy} - 2(\Delta x)(\Delta y) u_{xy} \right] \\
 &+ \frac{1}{3!} \left[-(\Delta x)^3 u_{xxx} + (\Delta y)^3 u_{yyy} + 3(\Delta x)^2(\Delta y) u_{xxy} - 3(\Delta x)(\Delta y)^2 u_{xyy} \right] \\
 &+ \frac{1}{4!} \left[(\Delta x)^4 u_{xxxx} + (\Delta y)^4 u_{yyyy} - 4(\Delta x)^3(\Delta y) u_{xxx} + 6(\Delta x)^2(\Delta y)^2 u_{xxyy} - 4(\Delta x)(\Delta y)^3 u_{xyyy} \right] \\
 &+ O[(\Delta x)^5, (\Delta y)^5]
 \end{aligned}$$

$$\begin{aligned}
 u_{i+1, j-1} &= u_{i,j} + (\Delta x) u_x - (\Delta y) u_y + \frac{1}{2!} \left[(\Delta x)^2 u_{xx} + (\Delta y)^2 u_{yy} - 2(\Delta x)(\Delta y) u_{xy} \right] \\
 &+ \frac{1}{3!} \left[(\Delta x)^3 u_{xxx} - (\Delta y)^3 u_{yyy} - 3(\Delta x)^2(\Delta y) u_{xxy} + 3(\Delta x)(\Delta y)^2 u_{xyy} \right] \\
 &+ \frac{1}{4!} \left[(\Delta x)^4 u_{xxxx} + (\Delta y)^4 u_{yyyy} - 4(\Delta x)^3(\Delta y) u_{xxx} + 6(\Delta x)^2(\Delta y)^2 u_{xxyy} - 4(\Delta x)(\Delta y)^3 u_{xyyy} \right] \\
 &+ O[(\Delta x)^5, (\Delta y)^5]
 \end{aligned}$$

$$\begin{aligned}
 &+ O[(\Delta x)^5, (\Delta y)^5] \\
 u_{i+1, j+1} - u_{i+1, j-1} - u_{i-1, j+1} + u_{i-1, j-1} &= \frac{1}{2!} \left[8(\Delta x)(\Delta y) u_{xy} \right] + \frac{1}{4!} \left[16(\Delta x)^3(\Delta y) u_{xxx} + 16(\Delta x)(\Delta y)^3 u_{xyy} \right]
 \end{aligned}$$

$$\therefore \frac{\partial^2 u(i,j)}{\partial x \partial y} \approx \frac{u_{i+1, j+1} - u_{i+1, j-1} - u_{i-1, j+1} + u_{i-1, j-1}}{4(\Delta x)(\Delta y)} - \left[\frac{1}{6} (\Delta x)^2 u_{xxx} + \frac{1}{6} (\Delta y)^2 u_{xyy} \right]$$

$$\therefore \text{the Leading-order Error is } \frac{1}{6} \left[(\Delta x)^2 \frac{\partial^3 u(i,j)}{\partial x^3 \partial y} + (\Delta y)^2 \frac{\partial^3 u(i,j)}{\partial x \partial y^3} \right]$$

$$b) u(x, y) = y^4 \sin x$$

$$\frac{\partial u}{\partial x} = y^4 \cos x$$

$$\frac{\partial u}{\partial y} = 4y^3 \sin x$$

$$\frac{\partial^2 u}{\partial x \partial y} = 4y^3 \cos x$$

Exact Value :

$$\therefore \frac{\partial^2}{\partial x \partial y} u(0, 1) = 4$$

$$u_{i+1, j+1} = u(0.1, 1.1) = 1.1^4 \sin(0.1) = 0.146166$$

$$u_{i+1, j-1} = u(0.1, 0.9) = 0.9^4 \sin(0.1) = 0.065501$$

$$u_{i-1, j+1} = u(-0.1, 1.1) = 1.1^4 \sin(-0.1) = -0.146166$$

$$u_{i-1, j-1} = u(-0.1, 0.9) = 0.9^4 \sin(-0.1) = -0.065501$$

Finite-difference Value :

$$\frac{\partial^2}{\partial x \partial y} u(0, 1) = \frac{u_{i+1, j+1} - u_{i+1, j-1} - u_{i-1, j+1} + u_{i-1, j-1}}{4(0.1)(0.1)} = 4.03325$$

c) The compute error in b) ≈ 0.03325

$$\text{The Leading-order error} = \frac{1}{6} \left[(\Delta x)^2 \frac{\partial^4 u}{\partial x^3 \partial y} + (\Delta y)^2 \frac{\partial^4 u}{\partial x \partial y^3} \right]$$

$$\therefore \frac{\partial^4 u}{\partial x^3 \partial y}(x, y) = -4y^3 \cos x \xrightarrow[y=1]{x=0} -4$$

$$\frac{\partial^4 u}{\partial x \partial y^3}(x, y) = 24y \cos x \xrightarrow[y=1]{x=0} 24$$

$$\therefore \frac{1}{6} \left[(0.1)^2 \frac{\partial^4 u}{\partial x^3 \partial y}(0, 1) + (0.1)^2 \frac{\partial^4 u}{\partial x \partial y^3}(0, 1) \right] = \frac{1}{6} \times 0.01 \times 20 = \frac{1}{30} \approx 0.0333$$

$\therefore$ Indeedly the error depend on the Leading-order error term.

Q10.

$$a) \frac{du}{dt} = \frac{10t}{t^5 + c} - \frac{25t^6}{(t^5 + c)^2} = \frac{2u}{t} - t^2 u^2$$

$$b) \frac{DF}{Dt} = \frac{\partial F}{\partial u} \frac{du}{dt} + \frac{\partial F}{\partial t}$$

$$F = \frac{du}{dt} = \frac{2u}{t} - t^2 u^2 \quad \therefore \frac{\partial F}{\partial u} = \frac{2}{t} - 2t^2 u \quad \frac{\partial F}{\partial t} = -\frac{2u}{t^2} - 2tu^2$$

$$\therefore \frac{DF}{Dt} = \left(\frac{2}{t} - 2t^2 u \right) \left(\frac{2u}{t} - t^2 u^2 \right) - \frac{2u}{t^2} - 2tu^2$$

$$= \frac{4u}{t^2} - 2tu^2 - 4tu^2 + 2t^4 u^3 - \frac{2u}{t^2} - 2tu^2$$

$$= \frac{2u}{t^2} - 8tu^2 + 2t^4 u^3$$

$$c) \frac{d^2 u}{dt^2} = \frac{d}{dt} \left(\frac{2u}{t} - t^2 u^2 \right) = -\frac{2u}{t^2} + \frac{2}{t} \frac{du}{dt} - 2tu^2 - t^2 \frac{du^2}{dt}$$

$$= -\frac{2u}{t^2} + \frac{2}{t} \left(\frac{du}{dt} \right) - 2tu^2 - 2ut^2 \left(\frac{du}{dt} \right)$$

$$= \frac{2u}{t^2} - 8tu^2 + 2t^4 u^3$$

$$\therefore \frac{DF}{Dt} = \frac{du}{dt}$$